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Cryo-EM structure of the bicarbonate receptor GPR30

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This study resolves a cryo-EM structure of the GPCR, human GPR30, which responds to bicarbonate and regulates cellular responses to pH and ion homeostasis. Understanding the ligand and the mechanism of activation is **important** to the field of receptor signaling and potentially facilitates drug development targeting this receptor. Structures and functional assays provide **solid** evidence for a potential bicarbonate binding site.

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Abstract

G-protein-coupled receptor 30 (GPR30) is a bicarbonate receptor that plays a vital role in cellular responses to extracellular pH and ion homeostasis. Despite its significance, the mechanisms by which GPR30 interacts with bicarbonate ions remain elusive. There is no consensus on a drug that targets GPR30, and difficulties in pharmacological analyses have limited biological and drug discovery research on GPR30. Here, we present the cryo-electron microscopy structure of human GPR30 in the presence of bicarbonate ions, at 3.15 Å resolution. Our structure reveals unique extracellular pockets and critical residues for bicarbonate binding and activation. Functional assays demonstrate that mutations in these residues impair bicarbonate-induced GPR30 activation, underscoring their importance in receptor function. This study also provides insights into G-protein coupling, highlighting the structural divergence between GPR30 and other GPCRs. Our findings not only advance the understanding of the role of GPR30 in pH homeostasis but also pave the way for the development of high-affinity drugs targeting GPR30 for therapeutic interventions in diseases associated with acid-base imbalance.

Introduction

The extracellular environment regulates cellular responses, as exemplified by pH and ion homeostasis. The acid-base balance, largely based on the bicarbonate buffer system *in vivo*, serves as a vital mechanism that maintains the optimal pH for cellular function.¹ Changes in the extracellular pH and ion homeostasis are monitored by membrane channels and receptors, such as G protein-coupled receptors (GPCRs). However, it remains unknown whether and how the acid-base balance-related ions, protons and bicarbonate ions, bind to the receptors and cause the conformational changes that lead to intracellular signal transduction. These proton-sensing GPCRs are thought to be involved in pH homeostasis, particularly in the acidic tumor microenvironment, at inflamed sites, and during ischemia-reperfusion injury.² In recent years, the determination of several proton receptor structures has begun to elucidate the molecular mechanisms underlying proton sensing.^{3–7}

We recently reported that a physiological concentration of bicarbonate ions, the counterpart of protons in the bicarbonate buffer system, activates G protein-coupled receptor 30 (GPR30), which leads to G_q -coupled calcium responses⁸. Our study also demonstrated that GPR30 in brain mural cells regulates blood flow and ischemia–reperfusion injury. GPR30 was identified as a G protein-coupled estrogen receptor that mediates the rapid non-genomic action of estradiol (E2)⁹. However, despite numerous reports on the pleiotropic functions of GPR30 *in vivo*^{10,11}, controversy remains regarding the responses of GPR30 to E2 *in vitro*¹², *ex vivo*¹³, and *in vivo*¹⁴. The broad expression of GPR30^{15,16}, including blood vessels^{17,18}, stomach, and lung, has also raised the possibility of its non-estrogenic functions. We recently demonstrated that three amino acids, E115^{2,61}, Q138^{3,33}, and H307^{7,36}, are essential for the bicarbonate-induced activation of GPR30, according to the public homology model (<https://gpcrdb.org/>)¹⁹. However, we did not examine whether bicarbonate ions interact with GPR30 and cause conformational changes. Moreover, to date, there is no consensus on a high-affinity drug that targets GPR30, and difficulties in pharmacological analyses have limited biological and drug discovery research on GPR30. To elucidate the bicarbonate–GPR30 interaction and to establish the first structural identification of an acid-base balance-related GPCR, we report the cryo-electron microscopy (cryo-EM) structure of human GPR30 in the presence of bicarbonate ions.

Results

Overall structure

For the structural study, we used the full-length human GPR30 sequence. To efficiently purify the stable GPCR-G-protein complex, the receptor and mini- G_{sqi} were incorporated in a ‘Fusion-G system’ by combining two complex stabilization techniques^{20,21} (Figure 1—figure supplement 1A, B). The modified receptor and G-protein were co-expressed in Sf9 cells and purified by Flag affinity chromatography in the presence of 200 mM NaHCO_3 . After an incubation with nanobody 35 (Nb35) and single chain scFv16, which binds to mini- G_{sqi} , the complex was purified by size exclusion chromatography (Figure 1—figure supplement 1C, D). The structure of the purified complex was determined by single-particle cryo-EM analysis, with an overall resolution of 3.15 Å (Table 1 [↗](#), Figure 1—figure supplement 2, and “Methods”). As the extracellular portion of the receptor was poorly resolved, we performed receptor focused refinement, yielding a density map with a nominal resolution of 3.18 Å, which was combined with the refined map focused on G-protein. The resulting composite map allowed us to precisely build the atomic model of the components, including the receptor (residues 51 to 196, 207 to 288, and 299 to 340, G-proteins, and scFv16) (Figure 1A, B [↗](#)). Nb35 was not visible in the cryo-EM map, probably because of the effect of the mini- G_s modification, although the interactive residues with Nb35 were not changed.

The receptor consists of the canonical 7 transmembrane helices (TM) connected by three intracellular loops (ICL1–3) and three extracellular loops (ECL1–3), and the amphipathic helix 8 at the C-terminus (H8) (Figure 2A [↗](#)). Most of the TMs are kinked, as in typical GPCRs²². It should be noted that TM1 is also kinked at P71^{1,44} (superscripts indicate Ballesteros-Weinstein numbers²³), whereas TM1 adopts a straight α -helix in most class A GPCRs. ICL1 and ECL1 contain short α helices (Figure 2B, C [↗](#)). The short ICL3 was completely visible in the cryo-EM map (Figure 2C [↗](#) and Figure 1—figure supplement 2), while residues 197 to 206 in ECL2 and 289 to 298 in ECL3 were disordered (Figure 2B [↗](#)). ECL2 is attached to TM3 by the disulfide bond between C130^{3,25} and C207^{ECL2} (Figure 2B [↗](#), Figure 2—figure supplement 1A–F), which is highly conserved in class A GPCRs^{22,24}. The cryo-EM structure did not superimpose well on the AlphaFold-predicted structure (Q99527 in the AlphaFold database)²⁵, with a root mean square deviation (R.M.S.D.) of Ca atoms of 3.10 Å (Figure 2—figure supplement 2A). The conserved D^{3,49}R^{3,50}Y^{3,51} motif in the predicted structure represents an inactive state (Figure 2—figure supplement 1B)²². Moreover, ECLs 2 and 3 are rich in cysteine residues, which form incorrect disulfide bonds in the predicted structure (Figure 2—figure supplement 2C). These comparisons support the usefulness of experimental structural determination.

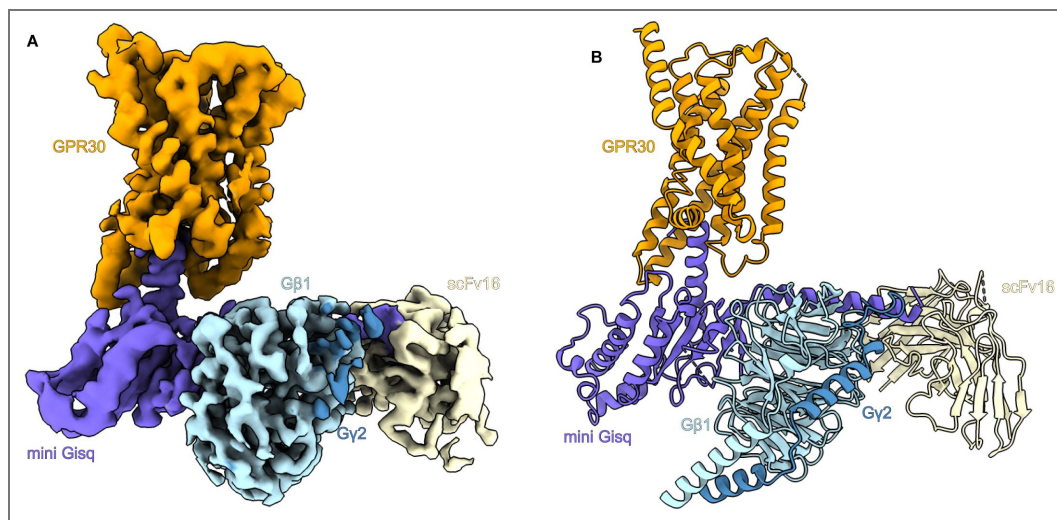


Figure 1. Overall structure of the GPR30-miniG_{sqi}β₁γ₂-scFv16 complex.

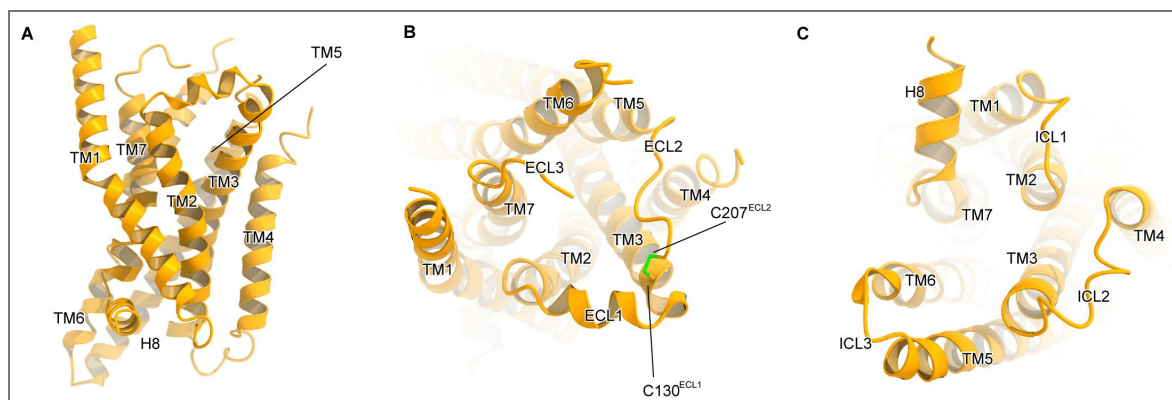
A Unsharpened cryo-EM density map of the GPR30-miniG_{sqi}β₁γ₂-scFv16-Nb35 complex, with the components individually colored. **B** The refined structure of the complex is shown as a ribbon representation.

Table 1. Cryo-EM data collection, refinement and validation statistics

Data collection	GPR30-G _i complex	
Microscope	Titan Krios (Thermo Fisher Scientific)	
Voltage (keV)	300	
Electron exposure (e ⁻ /Å ²)	50.660	
Detector	Gatan K3 Summit camera (Gatan)	
Magnification	×105,000	
Defocus range (μm)	-0.8–1.6	
Pixel size (Å/pix)	1.16	
Number of movies	9,824	
Symmetry	C1	
Picked particles	10,148,422	
Final particles	522,404	
Map resolution (Å)	3.15	
FSC threshold	0.143	
Model refinement		
Atoms		
R.m.s. deviations from ideal		
Bond lengths (Å)	0.003	
Bond angles (°)	0.528	
Validation		
Clashscore	9.56	
Rotamers Outliers(%)	0.23	
Ramachandran plot		
Favored (%)	96.32	
Allowed (%)	3.68	
Outlier (%)	0	

Figure 2. Receptor structure.

A–C Overall structure of the receptor, viewed from the membrane plane (**A**), intracellular side (**B**), and extracellular side (**C**).



Extracellular pockets

The interaction network between ECL1–3 covers the extracellular side of the receptor. As described above, ECL2 is anchored to TM3 via the disulfide bond, and ECL3 extends between ECL1 and ECL2, with R299^{ECL3} located within the receptor cavity (Figure 3A, B [↗](#)). R299^{ECL3} forms an electrostatic interaction with D210^{ECL2}. These interactions among the ECLs create three extracellular pockets (pockets A–C) (Figure 3A [↗](#)). Pocket B consists of ECL1, TM1, and TM7 (Figure 3C [↗](#)), while pocket C is formed by TM6, TM7, and ECL1 (Figure 3D [↗](#)). These two pockets are small, superficial, and hydrophobic and appear unsuitable for bicarbonate binding. Pocket A consists of ECL1–3 and TM2–7 and is the largest among the three pockets. It is connected to the inside of the receptor, and deep enough to reach W272^{6.48} (Figure 3B [↗](#)). Pocket A contains numerous hydrophilic residues favorable for bicarbonate binding (Figure 3E [↗](#)), suggesting that pocket A is a good candidate for the bicarbonate binding site.

However, no obvious density corresponding to bicarbonate is observed in pocket A, along with pockets B and C. In cryo-EM, negative charges are generally difficult to visualize, and the local resolution of the extracellular region in this structure is moderate (Figure 1—figure supplement 2). Thus, even if bicarbonate ions are present, it is possible that they cannot be visualized.

Our previous studies have shown that the mutations of E115^{2.61}, Q138^{3.33}, and H307^{7.36} abolish the bicarbonate-dependent calcium response of GPR30, suggesting their contribution to bicarbonate binding. To predict the bicarbonate binding site, we performed an exhaustive mutant analysis of the hydrophilic residues in pocket A, which was not done in the previous study⁸. We performed calcium assays using cell lines stably expressing N-terminally HA-tagged GPR30 and its mutants (Figure 3F, G [↗](#)), whose surface expression levels were analyzed by fluorescence-activated cell sorting (FACS) using an HA-antibody (Figure 3—figure supplement 1–3). Moreover, we performed TGF α shedding assays²⁶ using HEK cells transiently expressing the receptors (Figure 3—figure supplement 4A–F). The HA-tagged GPR30 wild-type showed a bicarbonate-dependent response. By contrast, the C207^{ECL2}A mutation that disrupts the disulfide bond between ECL2 and TM3, and the P71^{1.44} mutation that disrupts the characteristic kink in TM1, abolished the bicarbonate-dependent response. The expression levels of these mutants on the cell surface are significantly reduced (Figure 3—figure supplement 1, 4F), indicating that their reduced calcium responses are due to indirect effects such as structural instability, rather than the direct binding of bicarbonate ions. These findings underscore the critical role of the structural integrity of the receptor and provide compelling evidence that the bicarbonate response is mediated through GPR30.

Among the hydrophilic residue mutants, the D125^{ECL1}A mutant abolished the bicarbonate-induced responses (Figure 3F, G [↗](#), and Figure 3—figure supplement 4A, B). Q138^{3.33}A and D210^{ECL2}A reduced the responses and showed significant reductions in their cell surface expression levels (Figure 3F, G [↗](#), and Figure 3—figure supplement 1, 4A, B, F). However, caution is required in interpreting the results, since the cell surface expression level of the Q215^{5.39}A mutant was reduced to the same extent as Q138^{3.33}A and D210^{ECL2}A, but its bicarbonate-induced calcium response was as high as that of WT (Figure 3—figure supplement 1).

Combining these data with previous studies, we mapped the residues that cause decreased bicarbonate responses due to mutations on the current structure (P71^{1.44}, E115^{2.60}, D125^{ECL1}, Q138^{3.33}, C207^{ECL2}, D210^{ECL2}, and H307^{7.36}) (Figure 3E [↗](#)). Except for the structure-contributing residues (C207^{ECL2} and P71^{1.44}), the essential residues face pocket A, further supporting its role as the bicarbonate binding site. The D210^{ECL2}A mutant retained a moderate bicarbonate-induced calcium response (Figure 3F, G [↗](#)) and would contribute to the structural stability of the pocket through its interaction with R299^{ECL3} (Figure 3A [↗](#)). The D125^{ECL1}A mutant has lost its activity but is located on the surface (Figure 3E–G [↗](#)). Thus, D125^{ECL1} is unlikely to be a bicarbonate binding site, and the mutational effect could be explained due to the decreased surface expression (Figure 3—figure supplement 1, 4F). Given that most bicarbonate binding modes in other structures require recognition by positively charged residues, bicarbonate may bind to H307^{7.36}, while

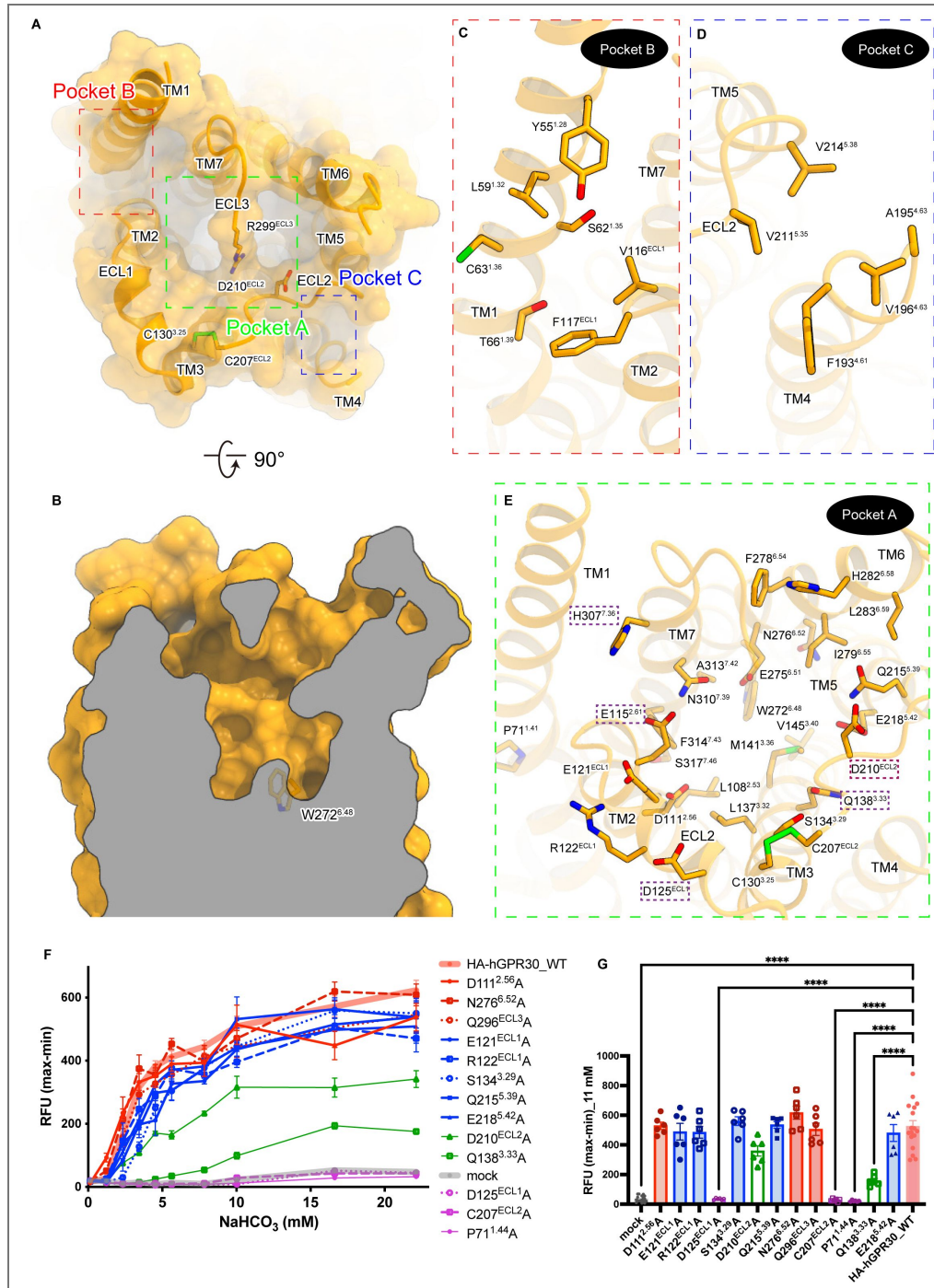


Figure 3. Architecture of the extracellular pocket.

A Molecular surface of the extracellular side. **B** Cross section of the pocket. **C–E** Residues facing pocket B (**C**), pocket C (**D**), and pocket A (**E**). In panel (**E**), only residues with reduced bicarbonate responses are highlighted. **F**, **G** Calcium assay using stable HEK293 cell lines expressing the N-terminal HA-tagged GPR30 wild-type (WT) and mutants. The mutants D111^{2.56}A, N276^{6.52}A, and Q296^{ECL3}A; E121^{ECL1}A, R122^{ECL1}A, S134^{3.29}A, Q215^{5.39}A, and E218^{5.42}A; D210^{ECL2}A and Q138^{3.33}A; and D125^{ECL1}A, C207^{ECL2}A, and P71^{1.44}A are highlighted in red, blue, green, and purple, respectively. The cells were stimulated by the indicated concentrations of NaHCO₃ at the timepoint of t = 20 sec. The Y axis indicates the difference between the maximum and minimum fluorescent values during 15 to 60 sec. Statistical analysis: each sample was compared with WT cells using Dunnett's T3 multiple comparisons test after Brown-Forsythe and Welch ANOVA tests. **** *p* < 0.0001 compared to WT cells, no significant difference is not shown. Data are presented as mean values ± SEM (**F**) and each replicate with the mean ± SEM (**G**).

E115^{2,61}, which is located near H307^{7,36}, may play an essential role by coordinating cations for bicarbonate binding, as observed in other structures. These residues are evolutionarily conserved from fish to human, further supporting their importance (Figure 3—figure supplement 5).

G-protein coupling

The C-terminal helix of Gα_q (α5-helix) deeply enters the intracellular cavity of GPR30, resulting in the formation of an active signaling complex. On the C-terminal side of the α5-helix, the backbone carbonyl of Y243^{G.H5.23} (superscript indicates the common Ga numbering [CGN] system²⁷) hydrogen bonds with R155^{3.50} in the conserved D^{3.49}R^{3.50}Y^{3.51} motif²² (Figure 4A). In addition, N244^{G.H5.24} forms a hydrogen bond with the peptide backbone of Y324^{7.53}. ICL3 faces the middle part of the α5-helix, with three characteristic arginine residues (R248^{6.24}, R251^{6.27}, and R254^{6.30}) surrounding D233^{G.H5.13} (Figure 4B). R248^{6.24} and R254^{6.30} form hydrogen bonds with D233^{G.H5.13} and Q237^{G.H5.17}, respectively. At ICL2, M163^{34.51} fits into a hydrophobic pocket composed of the α5-helix, the αN-β2 loop, and the β2-β3 loop of the Gα subunit, as in other GPCR-G_q complexes^{28–32} (Figure 4C–H). Above it, L159^{3.54} and A162^{34.50} form extensive hydrophobic interactions with L235^{G.H5.15}, L236^{G.H5.16}, and L240^{G.H5.20} in the α5-helix. These tight interactions with the α5-helix would enable the coupling of GPR30 with G_q.

The position of the α5-helix aligns with those in other GPCR-G_q complexes, but the position of the αN-helix is rotated away from the receptor (Figure 4I). The superimposition of the Gα subunits indicates that the rotation is not due to the movement of the entire G protein, but rather a 33° bend in the αN-helix (Figure 4J). This is attributed to variations in the interactions of the αN-helix with its receptors. The αN-helix interacts with the intracellular ends of ICL2 and TM4 in receptor-specific modes. In most G_q-coupled structures, ICL2 forms an α-helix and R^{34.55} interacts with R38^{G.hns1.03} at the end of the αN-helix. A typical example is the head-to-toe interaction between R185^{34.55} and R32^{G.hns1.03} in the serotonin 5-HT_{2A} receptor³¹ (Figure 4E). However, in GPR30, ICL2 does not adopt an α-helix and R^{34.55} is replaced by L167^{34.55}, which displaces R32^{G.hns1.03} downward through their interaction. Moreover, T170^{34.58} and R169^{34.57} form direct hydrogen bonds with R31^{G.hns1.02} and Y243^{G.H5.23}, respectively, which are not observed in other GPCR-G_q complexes. We hypothesize that these interactions may underpin the distinct αN rotation: A similar interaction between R38^{G.hns1.03} and L136^{34.54} is also observed in histamine H1 receptor (H₁R)²⁸ (Figure 4D), in which translational rather than rotational motion of αN is observed (Figure 4J).

Discussion

Our study has elucidated the distinctive structure of GPR30, which is unique among the class A GPCRs. The GPR30 structure is characterized by the kink of TM1 and the presence of the large extracellular pocket covered by ECLs. This pocket is rich in hydrophilic residues and could be a focus for the development of high-affinity drugs targeting GPR30. Our comprehensive mutagenesis study of the binding pocket complemented our previous study and mapped the essential residues for the bicarbonate-induced calcium response. Mutations in the S-S bond between ECL2 and TM3 and the bend in TM1 support their importance in the stability and the shape of the GPR30 structure and pocket. Furthermore, we were able to identify residues in pocket A that are important for the bicarbonate response, although there is room for debate regarding whether they directly contribute to binding or their mutations caused significant reductions in membrane expression. Considering the binding sites of negatively charged bicarbonate in other protein structures, it is reasonable to regard the region around H307^{7,36} as a bicarbonate binding site candidate. To clarify the binding mode of bicarbonate, a higher resolution structural analysis and complementary functional assessments are necessary.

GPR30 exhibits low homology to other GPCRs, with even closely related receptors sharing less than 30% sequence identity. A BLAST analysis revealed that GPR30 shares the highest sequence identity of 28% with the type 2 angiotensin II receptor (AT₂R) (Figure 5—figure supplement 1). Consequently, we compared the current structure of GPR30 with the ligand-bound AT₂R³³. The superimposition of GPR30 and AT₂R, with a Cα R.M.S.D. of 1.73 Å, suggests significant structural

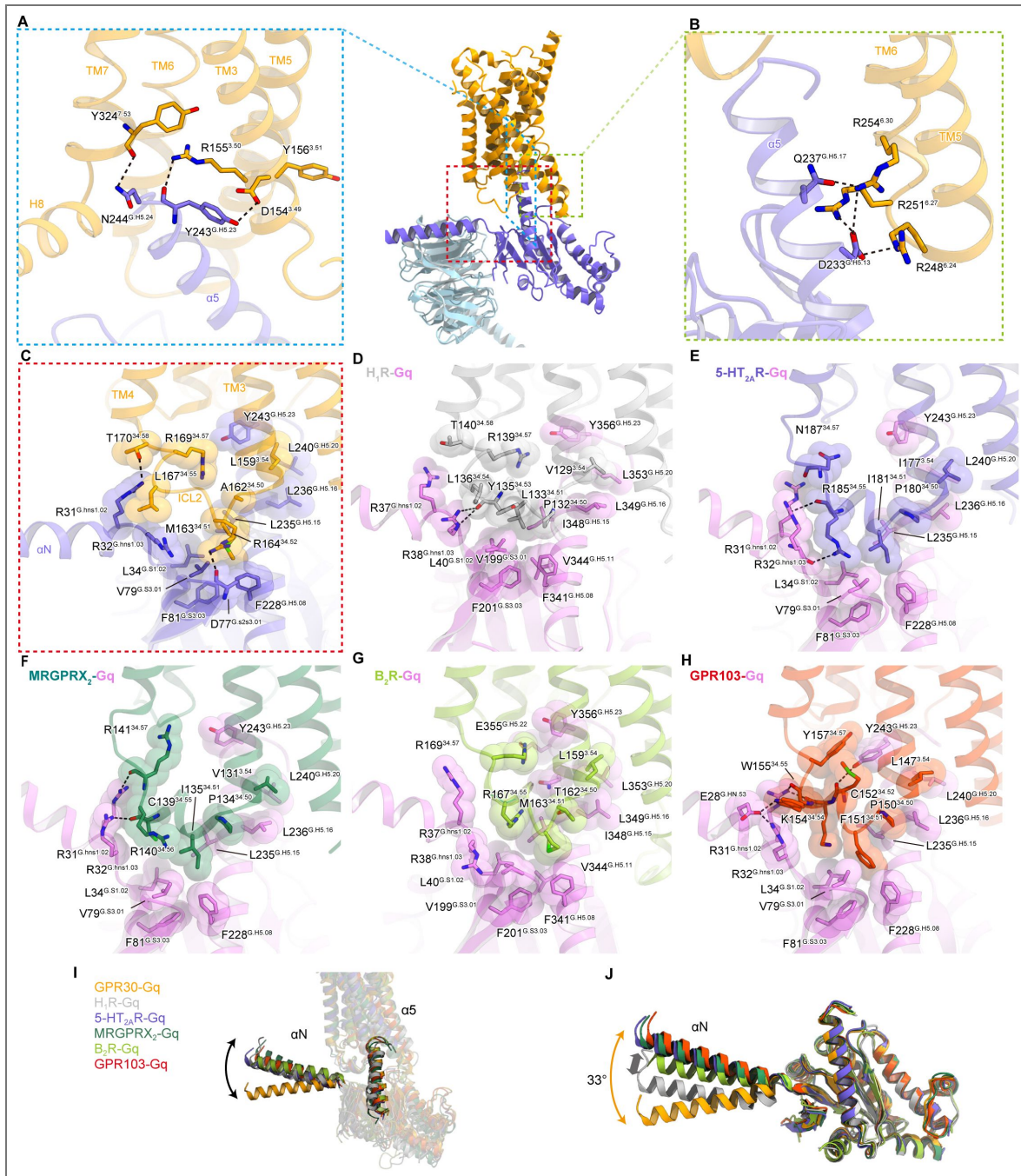


Figure 4. G-protein coupling.

A Hydrogen-bonding interactions between the C-terminal $\alpha 5$ -helix residues and the receptor. **B** Electrostatic and hydrogen-binding interactions between ICL3 and the $\alpha 5$ -helix. **C–H** Interface between ICL2 and G_q , with residues involved in hydrophobic interactions represented by CPK models. G-protein-bound GPCRs used in the comparison are as follows: H_1R - G_q (PDB 7DFL, gray), B_2R - G_q (PDB 7F6I, yellow-green), $MRGPRX_2$ - G_q (PDB 7S8L, green), $5-HT_{2A}$ - G_q (PDB 6WHA, gray), and $GPR103$ - G_q (PDB 8ZH8, red). **I** Comparison of the angles and positions of $\alpha 5h$ and αN relative to the receptor. **J** Superimposition of the G_a subunits.

divergence between the two (Figure 5A [↗](#)). Notably, there is no structural commonality between GPR30 and AT₂R in terms of the orientations of their respective TMs. Specifically, the extracellular half of TM1 in GPR30 extends outward, and is kinked at P71^{1.44} in the center of TM1 (Figure 5B [↗](#)). Near P71^{1.44}, Y65^{1.38} and F70^{1.43} are oriented toward the interior of the receptor, forming hydrophobic interactions with L113^{2.58} and L331^{7.40} to stabilize the kink. In the examination of the loops, ECL2 of AT₂R features a short β -sheet commonly found in peptide-activated class A GPCRs^{34–36}, whereas that of GPR30 adopts an elongated conformation that covers the extracellular pocket essential for the bicarbonate response (Figure 5A, B [↗](#)). These comparisons underscore the unique structural characteristics of GPR30.

In most GPCRs, TM1 adopts a straight α -helix and is positioned away from the orthosteric pocket formed by TM2–7. However, TM1 undergoes an inward movement due to allosteric conformational changes during agonist binding. The cannabinoid receptor CB₁ and adenosine A_{2A} receptor (A_{2A}R) exemplify this phenomenon, with TM1 kinked at G127^{1.43} and G23^{1.49} folding inward upon ligand binding^{37–40} (Figure 5C, D [↗](#)). By contrast, TM1 moves 4 Å outward in the endothelin ET_B receptor^{41,42} (Figure 5E [↗](#)). In GPR30, TM1 is kinked at P71^{1.44} (Figure 5B [↗](#)), indicating that this kink exists regardless of the receptor activation state. Indeed, the P71^{1.44}A mutation, which eliminates the kink in TM1, completely abolished the bicarbonate response (Figure 3F, G [↗](#)). Moreover, P71^{1.44} is evolutionarily conserved (Figure 3—figure supplement 5). These findings suggest the critical role of the TM1 kink in both the putative conformational changes associated with ligand binding and the functional integrity of GPR30.

During the revision of this manuscript, the structures of apo-GPR30-G_q (PDB 8XOG) and the exogenous ligand Lys05-bound GPR30-G_q (PDB 8XOF) were reported⁴³. We compared our structure of GPR30 in the presence of bicarbonate with these structures. In the extracellular region, the position of TM5 in GPR30 in the presence of bicarbonate is similar to that in apo-GPR30. In contrast, the position of TM6 is shifted outward relative to that of apo-GPR30, resembling the conformation observed in Lys05-bound GPR30 (Figure 6A, B [↗](#)). Additionally, the position of ECL1 is also shifted outward compared to that of apo-GPR30 (Figure 6B [↗](#)). In the GPR30 structure in the presence of bicarbonate, ECL2 was modeled, suggesting differences in structural flexibility. These findings indicate that the structure of GPR30 in the presence of bicarbonate is different from both the apo structure and the Lys05-bound structure, demonstrating that the structure and the flexibility of the extracellular domain of GPR30 change depending on the type of ligand. Furthermore, focusing on the interaction with G_q, the α N helix of G_q is not rotated in the structure bound to Lys05, in contrast to the characteristic bending of the α N helix in our structure (Figure 6C, D [↗](#)). Although it is necessary to consider variations in experimental conditions, such as salt concentration, the differences in the G_q binding modes suggest that the downstream signals may change in a ligand-dependent manner.

In our work, we used relatively high (mM) concentrations of bicarbonate to activate GPR30. In parallel with its low potency, the physiological concentrations of bicarbonate are 22–26 mM in the extracellular fluid, including interstitial fluid and blood, and 10–12 mM in the cytoplasm. Therefore, GPR30 is activated by the physiological concentrations of bicarbonate in tissues. The bicarbonate concentrations are altered in various physiological and pathological conditions – metabolic acidosis in chronic kidney disease causes a drop to 2–3 mM, and metabolic alkalosis induced by severe emesis increases bicarbonate concentrations to over 30 mM. Thus, our work clearly shows that GPR30 is activated by physiological concentrations of bicarbonate, present both intracellularly and extracellularly, and that GPR30 can be deactivated or reactivated under various pathophysiological conditions. Thus, we have identified GPR30 as a bicarbonate receptor.

Methods

Expression and purification of scFv16 and Nb35

The gene encoding scFv16 was synthesized (GeneArt) and subcloned into a modified pFastBac vector⁴⁴, with the resulting construct encoding the GP67 secretion signal sequence at the N terminus, and a His₈ tag followed by a TEV cleavage site at the C terminus. The His₈-tagged scFv16

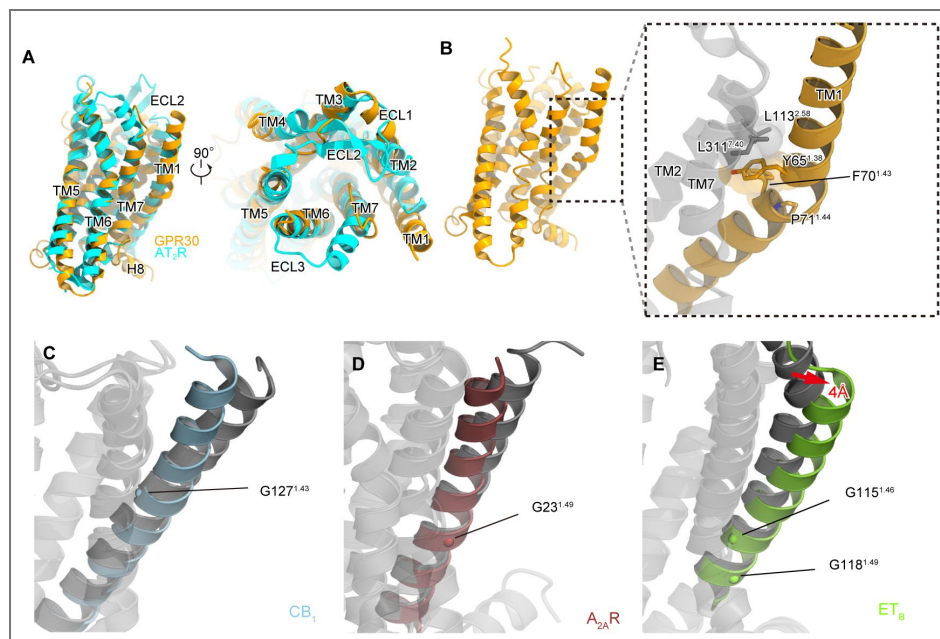


Figure 5. Structural comparison with related GPCRs.

A Structural comparison of GPR30 and AT₂R (PDB 5UNF). **B** Interactions around P71^{1.44} in TM1 of GPR30. **C–F** Conformational changes of TM1 upon agonist binding in CB₁ (**C**), A_{2A}R (**D**), and ET_B (**E**). The agonist-bound states are colored with the respective colors, while the inactive states are colored gray. The PDB codes used in this figure are CB₁-active (PDB 5XRA), CB₁-inactive (PDB 5TGZ), A_{2A}R-active (PDB 6GDG), A_{2A}R-inactive (PDB 3EML), ET_B-active (PDB 8IY5), and ET_B-inactive (PDB 5X93).

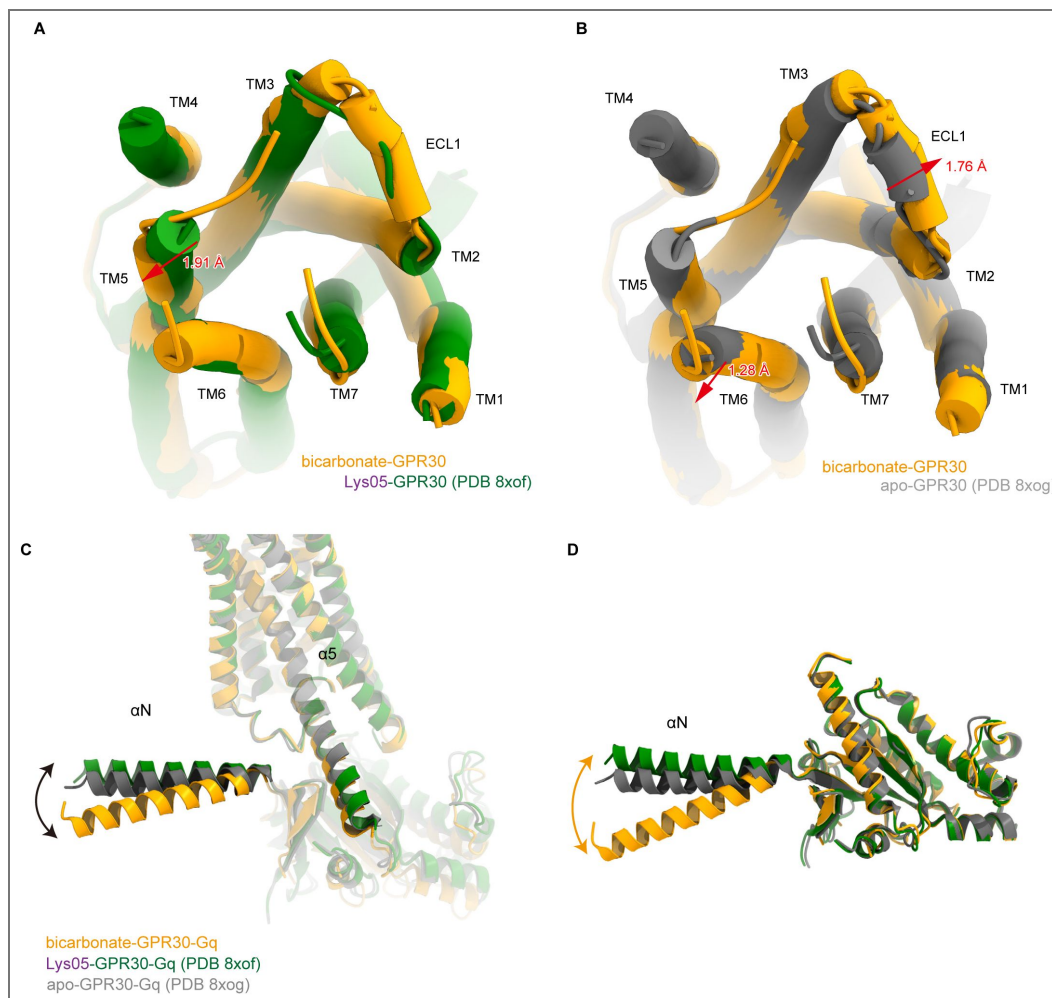


Figure 6. Structural comparison with GPR30 bound to Lys05.

A, B Superimposition of bicarbonate-GPR30, Lys05-binding GPR30 (PDB 8XOF) (**A**), and apo-GPR30 (PDB 8XOG) (**B**) structures, viewed from the extracellular side. **C** Comparison of the angles and positions of α5h and αN relative to the receptor. **D** Superimposition of the Ga subunits.

was expressed and secreted by Sf9 insect cells, as previously reported⁴⁵. The Sf9 cells were removed by centrifugation at 5,000g for 10 min, and the secreta-containing supernatant was combined with 5 mM CaCl₂, 1 mM NiCl₂, 20 mM HEPES (pH 8.0), and 150 mM NaCl. The supernatant was mixed with Ni Superflow resin (GE Healthcare Life Sciences) and stirred for 1 h at 4 °C. The collected resin was washed with buffer containing 20 mM HEPES (pH 8.0), 500 mM NaCl and 20 mM imidazole, and further washed with 10 column volumes of buffer containing 20 mM HEPES (pH 8.0), 500 mM NaCl and 20 mM imidazole. Next, the protein was eluted with 20 mM Tris (pH 8.0), 500 mM NaCl and 400 mM imidazole. The eluted fraction was concentrated and loaded onto a Superdex200 10/300 Increase size-exclusion column, equilibrated in buffer containing 20 mM Tris (pH 8.0) and 150 mM NaCl. Peak fractions were pooled, concentrated to 5 mg/ml using a centrifugal filter device (Millipore 10KDaCMW cutoff), and frozen in liquid nitrogen.

Nb35 was prepared as previously reported^{46,47}. In brief, Nb35 was expressed in the periplasm of *E. coli*. The harvested cells were disrupted by sonication. Nb35 was purified by nickel affinity chromatography, followed by gel-filtration chromatography, and frozen in liquid nitrogen.

Constructs for expression of GPR30 and G_q

The human GPR30 gene (UniProtKB, Q99527) was subcloned into a modified pFastBac vector²⁰, with an N-terminal haemagglutinin signal peptide followed by the Flag-tag epitope (DYKDDDDK) and the LgBiT fused to its C-terminus followed by a 3 C protease site and EGFP-His₈ tag. A 15 amino sequence of GGSGGGSGSSSGG was inserted into both the N-terminal and C-terminal sides of LgBiT. Rat Gβ₁ and bovine Gγ₂ were subcloned into the pFastBac Dual vector. In detail, rat Gβ₁ was cloned with a C-terminal HiBiT connected with a 15 amino sequence of GGSGGGSGSSSGG. Moreover, mini-G_{sqi} was subcloned into the C-terminus of the bovine Gγ₂ with a nine amino sequence GSAGSAGSA linker. The resulting pFastBac dual vector can express the mini-G_{sqi} trimer.

Expression and purification of the human GPR30 – G_q complex

The recombinant baculovirus was prepared using the Bac-to-Bac baculovirus expression system (Thermo Fisher Scientific). For expression, 2 L of Sf9 cells at a density of 3 × 10⁶ cells/mL were co-infected with baculovirus encoding GPR30 and mini-G_{sqi} trimer at the ratio of 1:1 and the cells were incubated at 30 °C. After 48 hours, the collected cells were resuspended and dounce-homogenized in 20 mM Tris-HCl, pH 8.0 (4 °C), 200 mM NaCl, 10% Glycerol, 200 mM NaHCO₃, 25 mU/ml apyrase. The crude membrane fraction was collected by ultracentrifugation at 180,000g for 1 h and solubilized in buffer, containing 20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1% n-dodecyl-beta-D-maltopyranoside (DDM) (Calbiochem), 0.2 % cholesteryl hemisuccinate (CHS) (Merck), 10% glycerol, 200 mM NaHCO₃, and 25 mU/ml Apyrase, for 1 h at 4 °C. The supernatant was separated from the insoluble material by ultracentrifugation at 180,000g for 20 min and incubated with 5 mL of M2 anti-flag affinity resin (Sigma) for 1 h at 4 °C. The resin was washed with 20 column volumes of buffer containing 20 mM Tris-HCl, pH 8.0 500 mM NaCl, 10% Glycerol, 0.05% Lauryl Maltose Neopentyl Glycol (LMNG) (Antrace), 0.005% CHS and 200 mM NaHCO₃. The complex was eluted in buffer containing 20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 10% Glycerol, 0.01% LMNG, 0.001% CHS, 200 mM NaHCO₃ and 0.2 mg/mL Flag peptide. The eluate was incubated with the Nb35 and scFv16 at 4 °C. The complex was concentrated and purified by size exclusion chromatography on a Superose 6 increase (GE) column in 20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 0.01% LMNG, 0.001% CHS and 200 mM NaHCO₃. Peak fractions were concentrated to 4.72 mg/ml for electron microscopy studies.

Sample vitrification and cryo-EM data acquisition

The purified complex was applied onto a freshly glow-discharged Quantifoil Au grid (R1.2/1.3, 300 mesh), and plunge-frozen in liquid ethane by using a Vitrobot Mark IV. Data collections were performed on a 300kV Titan Krios G3i microscope (Thermo Fisher Scientific) equipped with a BioQuantum K3 imaging filter and a K3 direct electron detector (Gatan).

A total of 9824 movies were acquired with a calibrated pixel size of $0.83\text{\AA}\text{Cpix}^{-1}$ and with a defocus range of -0.8 to $-1.6\text{\mu m}$, using EPU. Each movie was acquired for 2.6Cs and split into 48 frames, resulting in an accumulated exposure of about $50.660\text{Celectrons per \AA}^2$.

Image processing

All acquired dose-fractionated movies were imported into CryoSPARC v4.4⁴⁸ and subjected to beam-induced motion correction. The contrast transfer function (CTF) parameters were estimated and a total of 10,148,422 particles were extracted. The particles were subjected to 2D classifications, Ab-initio reconstruction and several rounds of hetero refinement and Non-uniform refinement. Next, the particles were subjected to 3D classification with a mask on the receptor. Then the particle sets were subjected to Reference Based Motion Correction. Motion-corrected 522,404 particles were subjected to Non-uniform refinement, yielding a map with a global nominal resolution of $3.15\text{\AA}$, with the gold standard Fourier Shell Correlation (FSC= 0.143) criteria⁴⁹. Moreover, the 3D model was refined with a mask on the receptor. As a result, the receptor has a higher resolution with a nominal resolution of $3.18\text{\AA}$. The overall and receptor focused maps were combined by phenix⁵⁰. The processing strategy is described in Supplementary Figure 2.

Model building and refinement

The density map was sharpened by phenix.auto_sharpen⁵¹ and the quality of the density map was sufficient to build a model manually in COOT^{52,53}. The model building was facilitated by the Alphafold-predicted structure. We manually fitted GPR30, the G_q heterotrimer and scFv16 into the map. We then manually readjusted the model into using COOT and refined it using phenix.real_space_refine^{50,54} (v.1.19) with the secondary-structure restraints using phenix.secondary_structure_restraints.

Vector construction and transfection

Human *Gpr30* cDNA was obtained from human hepatoblastoma-derived HepG2 cells. The coding sequences of human *Gpr30* were inserted into the multi-cloning site of the plasmid vector pCXN2, which was generated in our laboratory via modification of pCAGGS, between the KpnI and EcoRI sites. The N- and C-terminal HA-tagged *Gpr30* was amplified using a reverse primer containing the HA sequence. One amino acid mutation of human GPR30 was generated as follows: the targeted amino acid was changed to alanine (GCC) using a two-step PCR method with the QuikChange® Primer Design Program by Agilent (<https://www.agilent.com/store/primerDesignProgram.jsp> ) and the coding sequence with each mutation was inserted into the multi-cloning site of pCXN2 between the KpnI and EcoRI sites. The mutations generated were D111A, E121A, R122A, D125A, S134A, Q138A, D210A, Q215A, E218A, N276A, Q296A, C207A, P71A, and H307A.

Cell line sources and transfection

HEK293A cells (female origin; Thermo Fisher Scientific) were maintained at $37\text{ }^{\circ}\text{C}$ and 5% CO_2 . These vectors were transfected using the lipofection method (Lipofectamine™ 2000 Transfection Reagent, 11668019, Invitrogen). Stable cell lines were established through antibiotic drug selection. The HEK293A cells were transfected with the N-terminal HA-tagged wild-type and mutant *Gpr30* and maintained in the culture medium containing 1 mg/ml of G418. After 2-3 weeks of antibiotic drug selection, stable expression of exogenous GPR30 was confirmed by three independent flow cytometric analyses.

TGF α shedding assay

The TGF α shedding assay was performed according to a previously published protocol²⁶. The AP-TGF α expression vector was provided by Dr. Inoue and Dr. Aoki, Tohoku University. HEK293 cells were seeded in 12-well plates at a density of 1×10^5 cells/well and cultured for 24 h. At 70% confluency, a mixture of plasmid vectors containing GPCR (see '**Vector construction and transfection**' section) and AP-TGF α was transfected into the cells using Lipofectamine 2000 transfection reagent. After another 24 h of incubation, the cells were detached with 0.05%

trypsin/EDTA (32777-44, Nacalai Tesque), suspended in Hanks' balanced salt solution, and seeded in 96-well plates. The cells were stimulated for 1 h with $1\text{--}44\times 10^{-3}$ M (final concentration) of NaHCO_3 at 37°C , under 0.03% CO_2 . Conditioned media (CM) was transferred to another plate, and 80 μl of alkaline phosphatase (AP) solution (40 mM Tris-HCl, pH 9.5, 40 mM NaCl, 10 mM MgCl_2 , 10 mM p-nitrophenylphosphate disodium salt hexahydrate) was added to both plates, which were then incubated at 37°C . The optical density at 405 nm (OD405) was measured using a microplate reader (iMark, Bio-Rad) at 5 min, 30 min, 1 h, and 2 h, depending on the reaction rate. The percentage of shed AP-TGF was calculated using the following equations:

$$\text{AP activity} = \Delta\text{OD405} (1 - 0 \text{ h})$$

$$\% \text{ CM (conditioned media)} = \text{AP}_{\text{CM}} / (\text{AP}_{\text{CM}} + \text{AP}_{\text{Cell}})$$

Calcium assay

Stable HEK293 cells (2×10^4 cells/well) were seeded into a black wall and clear bottom 96-well plate 24 h before the assay. Cells at 90–100% confluency were incubated with HEPES buffer ($1\times$ HBSS, 2.5 mM probenecid, 25 mM HEPES, pH 7.4) containing 10 μM Fluo-8 AM (21080, AAT-Bio) for 60 min at 37°C and 5% CO_2 . The cells were washed twice with HEPES buffer. The cells were stimulated with indicated concentrations of NaHCO_3 and ATP at 37°C and 0.03% CO_2 . The fluorescence intensity was analysed using FlexStation 3 (Molecular Devices, Ex/Em = 490/525 nm).

Flow cytometry

HEK293A cells stably expressing the N-terminal HA-tagged wild-type and mutant GPR30 were incubated with the calcium-free Minimum Essential Medium (S-MEM, #11380037, Gibco™) for 30 min at 37°C and 5.0% CO_2 before the assay. The cells were blocked with PBS containing 2% goat serum for 10 min at 4°C , incubated with the biotin-conjugated anti-HA antibody (2.5 $\mu\text{g}/\text{ml}$, #12158167001, Roche) for 60 min at 4°C , and then stained with Alexa Fluor 488-conjugated anti-rat IgG (5 $\mu\text{g}/\text{ml}$, #A-11006, Thermo Fisher Scientific) for 40 min at 4°C . LSRFortessa (BD Biosciences) was used for analysis. For intracellular staining, the cells were fixed and stained using eBioscience™ Fixable Viability Dye eFluor 780 (#65-0865-14, Thermo Fisher Scientific) and eBioscience™ Fox3/Transcription Factor Staining Buffer Set (#00-5523-00, Thermo Fisher Scientific).

Analysis of cell surface expression of GPR30

The cell surface expression of wild-type and mutant hGPR30-HA was analysed via cell surface protein isolation using a Cell Surface Protein Isolation Kit (#89881, Thermo Scientific™) followed by western blotting. Briefly, HEK293 cells transiently expressing wild-type or mutant hGPR30-HA were biotinylated for 30 min at 4°C . Then, the cells were harvested, and biotinylated proteins were isolated with avidin binding.

Western blotting

Cell lysates were separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis under reducing conditions and transferred to polyvinylidene difluoride membranes (Immobilon P, IPVH00010, Millipore). Primary antibodies used were anti-HA High Affinity, 1:1000 dilution, 11867423001, Roche; Na-K-ATPase, 1:1000 dilution, #3010, Cell Signaling Technology; β -Actin (AC-15), 1:1000 dilution, sc-69879, Santa Cruz Biotechnology. Secondary antibodies used were anti-rabbit IgG, HRP-linked Antibody, 1:5000 dilution, #7074, Cell Signaling Technology; anti-mouse IgG, HRP-linked Antibody, 1:5000 dilution, #7076, Cell Signaling Technology; anti-rat IgG, HRP-Linked Whole Ab Goat, 1:5000 dilution, NA935, Cytiva. The membranes were probed at 4°C overnight with the primary antibodies. The membranes were subsequently incubated with the corresponding secondary antibodies. The signals were detected with ECL Prime (RPN2236, Cytiva) or ImmunoStar LD (296-69901, FUJIFILM Wako Pure Chemical Corporation) using a chemiluminescence imaging system (CFusion FX7, Vilber).

Data availability

The cryo-EM density map and atomic coordinates for the GPR30-Gq complex have been deposited in the Electron Microscopy Data Bank and the PDB, under the accession codes: EMD-66632 [<https://www.ebi.ac.uk/emdb/entry/EMD-66632>] (GPR30-Gq), EMD-66629 [<https://www.ebi.ac.uk/emdb/entry/EMD-66629>] (GPR30-Gq, consensus map), EMD-66630 [<https://www.ebi.ac.uk/emdb/entry/EMD-66630>] (GPR30-Gq, G-protein focused), EMD-66631 [<https://www.ebi.ac.uk/emdb/entry/EMD-66631>] (GPR30-Gq, receptor focused), and PDB 9X74 [<http://doi.org/10.2210/pdb9X74/pdb>]. Source data are provided with this paper. All other data are available from the corresponding authors upon reasonable request.

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Additional information

Author contributions

S.K. performed all the experiments involved in the structural determination, assisted by H.A. and H.S.O. W.S. designed the project and initially tried the expression of GPR30. A.J.-W. and T.Y. performed and oversaw the mutagenesis study. The manuscript was mainly prepared by S.K., H.A., and W.S., with assistance from O.N.

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Additional files

[Supplements](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study resolves a cryo-EM structure of the GPCR, GPR30, in the presence of bicarbonate, which the author's lab recently identified as the physiological ligand. Understanding the ligand and the mechanism of activation is of fundamental importance to the field of receptor signaling. This solid study provides important insight into the overall structure and suggests a possible bicarbonate binding site.

Strengths:

The overall structure, and proposed mechanism of G-protein coupling are solid. Based on the structure, the authors identify a binding pocket that might accommodate bicarbonate. Although assignment of the binding pocket is speculative, extensive mutagenesis of residues in this pocket identifies several that are important to G-protein signaling. The structure shows some conformational differences with a previous structure of this protein determined in the absence of bicarbonate (PMC11217264). To my knowledge, bicarbonate is the only physiological ligand that has been identified for GPR30, making this study an important contribution to the field. However, the current study provides novel and important circumstantial evidence for the bicarbonate binding site based on mutagenesis and functional assays.

Weaknesses:

Bicarbonate is a challenging ligand for structural and biochemical studies, and because of experimental limitations, this study does not elucidate the exact binding site. Higher resolution structures would be required for structural identification of bicarbonate. The functional assay monitors activation of GPR30, and thus reports on not only bicarbonate binding, but also the integrity of the allosteric network that transduces the binding signal

across the membrane. However, biochemical binding assays are challenging because the binding constant is weak, in the mM range.

The authors appropriately acknowledge the limitations of these experimental approaches, and they build a solid circumstantial case for the bicarbonate binding pocket based on extensive mutagenesis and functional analysis. However, the study does fall short of establishing the bicarbonate binding site.

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Reviewer #2 (Public review):

Summary:

In this manuscript, "Cryo-EM structure of the bicarbonate receptor GPR30," the authors aimed to enrich our understanding of the role of GPR30 in pH homeostasis by combining structural analysis with a receptor function assay. This work is a natural development and extension of their previous work on Nature Communications (PMID: 38413581). In the current body of work, they solved the cryo-EM structure of the human GPR30-G-protein (mini-Gsqi) complex in the presence of bicarbonate ions at 3.15 Å resolution. From the atomic model built based on this map, they observed the overall canonical architecture of class A GPCR and also identified 3 extracellular pockets created by ECLs (Pockets A-C). Based on the polarity, location, size, and charge of each pocket, the authors hypothesized that pocket A is a good candidate for the bicarbonate binding site. To identify the bicarbonate binding site, the authors performed an exhaustive mutant analysis of the hydrophilic residues in Pocket A and analyzed receptor reactivity via calcium assay. In addition, the human GPR30-G-protein complex model also enabled the authors to elucidate the G-protein coupling mechanism of this special class A GPCR, which plays a crucial role in pH homeostasis.

Strengths:

As a continuation of their recent Nature Communications publication, the authors used cryo-EM coupled with mutagenesis and functional studies to elucidate bicarbonate-GPR30 interaction. This work provided atomic-resolution structural observations for the receptor in complex with G-protein, allowing us to explore its mechanism of action, and will further facilitate drug development targeting GPR30. There were 3 extracellular pockets created by ECLs (Pockets A-C). The authors were able to filter out 2 of them and hypothesized that pocket A was a good candidate for the bicarbonate binding site based on the polarity, location, and charge of each pocket. From there, the authors identified the key residues on GPR30 for its interaction with the substrate, bicarbonate. Together with their previous work, they mapped out amino acids that are critical for receptor reactivity.

Weaknesses:

When we see a reduction of a GPCR-mediated downstream signaling, several factors could potentially contribute to this observation: 1) a reduced total expression of this receptor due to the mutation (transcription and translation issue); 2) a reduced surface expression of this receptor due to the mutation (trafficking issue); and 3) a dysfunctional receptor that doesn't signal due to the mutation. In the current revision, based on the gating strategy, the surface expression of the HA-positive WT GPR30-expressing cells is only 10.6% of the total population, while the surface expression levels of the mutants range from 1.89% (P71A) to 64.4% (D111A). Combining this information with the functional readout in Figure 3F and G, as well as their previous work, the authors concluded that mutations at P71, E115, D125, Q138, C207, D210, and H307 would decrease bicarbonate responses. Among those sites,

E115, Q138, and H307 were from their previous Nature Comm paper.

Authors claim P71 and C207 make a structural-stability contribution, as their mutations result in a significant reduction in surface expression: P71A (1.89%) and C207A (2.71%). However, compared to 10.6% of the total population in the WT, (P71A is 17.8% of the WT, and C207A is 25.6% of the WT), this doesn't rule out the possibility that the mutated receptor is also dysfunctional: at 10 mM NaHCO₃, RFU of WT is ~500, RFU of P71 and C207 are ~0.

The authors also interpret "The D125ECL1A mutant has lost its activity but is located on the surface" and only mention "D125 is unlikely to be a bicarbonate binding site, and the mutational effect could be explained due to the decreased surface expression". Again, compared to 10.6% of the total population in the WT, D125A (3.94%) is 37.2% of the WT. At 10 mM NaHCO₃, the RFU of the WT is ~500, the RFU of D125 is ~0. This doesn't rule out the possibility that the mutated receptor is also dysfunctional. It is not clear why D125A didn't make it to the surface.

Other mutants that the authors didn't mention much in their text: D111A (64.4%, 607.5% of WT surface expression), E121A (50.4%, 475.5% of WT surface expression), R122 (41.0%, 386.8% of WT surface expression), N276A (38.9%, 367.0% of WT surface expression) and E218A (24.6%, 232.1% of WT surface expression) all have similar RFU as WT, although the surface expression is about 2-6 times more. On the other hand, Q215A (3.18%, 30% of WT surface expression) has similar RFU as WT, with only a third of the receptor on the surface.

Altogether, the wide range of surface expression across the different cell lines, combined with the different receptor function readouts, makes the cell functional data only partially support their structural observations.

<https://doi.org/10.7554/eLife.99874.2.sa2>

Reviewer #3 (Public review):

Summary

GPR30 responds to bicarbonate and plays a role in regulating cellular pH and ion homeostasis. However, the molecular basis of bicarbonate recognition by GPR30 remains unresolved. This study reports the cryo-EM structure of GPR30 bound to a chimeric mini-Gq in the presence of bicarbonate, revealing mechanistic insights into its G-protein coupling. Nonetheless, the study does not identify the bicarbonate-binding site within GPR30.

Strengths

The work provides strong structural evidence clarifying how GPR30 engages and couples with Gq.

Weaknesses

Several GPR30 mutants exhibited diminished responses to bicarbonate, but their expression levels were also reduced. As a result, the mechanism by which GPR30 recognizes bicarbonate remains uncertain, leaving this aspect of the study incomplete.

<https://doi.org/10.7554/eLife.99874.2.sa1>

Author response:

The following is the authors' response to the original reviews.

The parts of the text that have been changed. The major changes are as follows:

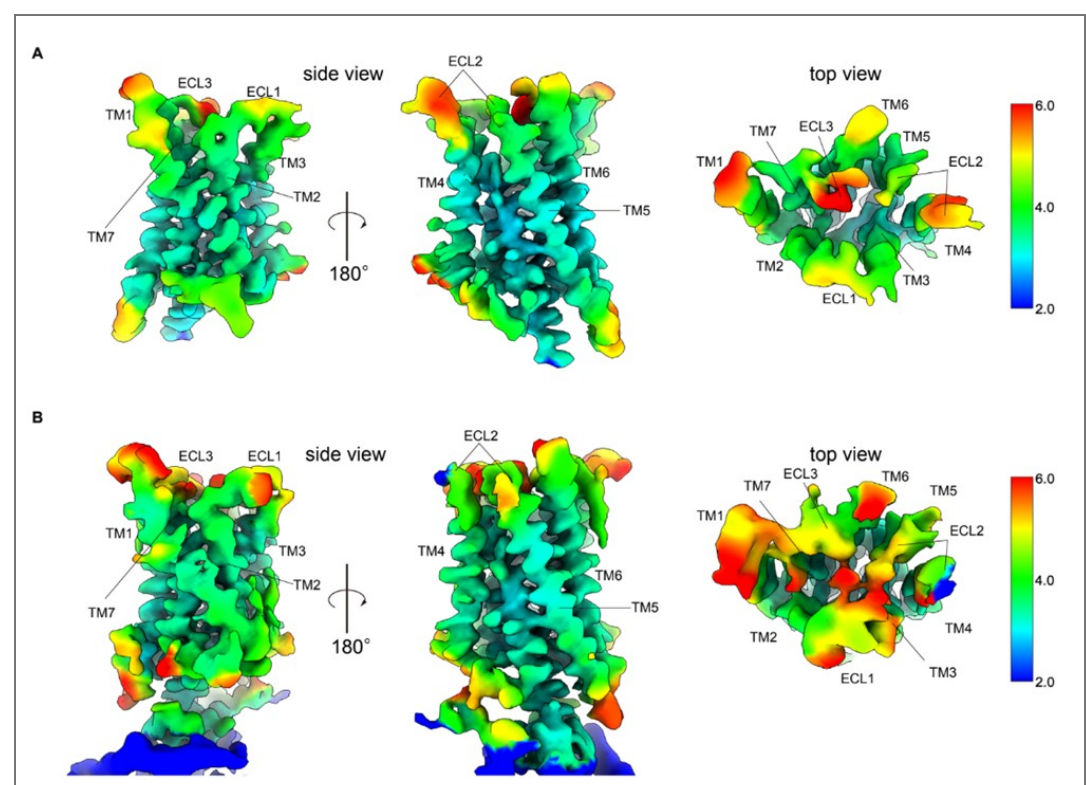
We re-analyzed the dataset and improved the local resolution of the extracellular region (Author response image 1).

We re-modeled based on the improved density and canceled the bicarbonate model based on comments from all reviewers.

We performed calcium assay using cell lines stably expressing the mutants, whose surface expression levels were analyzed by fluorescence-activated cell sorting (FACS)

(Figure 3F, G and Figure 3–figure supplement 1-3).

Thus, we significantly revised our discussion of the extracellular binding pocket and the result of the mutational study. In the revised manuscript, we speculate that H307 is a candidate for the bicarbonate binding site.



Author response image 1. Figure Comparison of local resolution between re-analyzed and previous maps. A

Side and top view of the re-analyzed receptor-focused map of GPR30 colored by local resolution. B Side and top view of the previous receptor-focused map of GPR30 colored by local resolution

Reviewer #1 (Public Review):

Summary:

This study resolves a cryo-EM structure of the GPCR, GPR30, which was recently identified as a bicarbonate receptor by the authors' lab. Understanding the ligand and the mechanism of activation is of fundamental importance to the field of receptor signaling. However, the main claim of the paper, the identification of the bicarbonate binding site, is only partly supported by the structural and functional data, leaving the study incomplete.

Strengths:

The overall structure, and proposed mechanism of G-protein coupling seem solid. The authors perform fairly extensive unbiased mutagenesis to identify a host of positions that are important to G-protein signaling. To my knowledge, bicarbonate is the only physiological ligand that has been identified for GPR30, making this study a particularly important contribution to the field.

Weaknesses:

Without higher resolution structures and/or additional experimental assessment of the binding pocket, the assignment of the bicarbonate remains highly speculative. The local resolution is especially poor in the ECL loop region where the ligand is proposed to bind (4.3 - 4.8 Å range). Of course, sometimes it is difficult to achieve high structural resolution, but in these cases, the assignment of ligands should be backed up by even more rigorous experimental validation. The functional assay monitors activation of GPR30, and thus reports on not only bicarbonate binding, but also the integrity of the allosteric network that transduces the binding signal across the membrane. Thus, disruption of bicarbonate signaling by mutagenesis of the putative coordinating residues does not necessarily mean that bicarbonate binding has been disrupted. Moreover, the mutagenesis was apparently done prior to structure determination, meaning that residues proposed to directly surround bicarbonate binding, such as E218, were not experimentally validated. Targeted mutagenesis based on the structure would strengthen the story.

Moreover, the proposed bicarbonate binding site is surprising in a chemical sense, as it is located within an acidic pocket. The authors cite several other structural studies to support the surprising observation of anionic bicarbonate surrounded by glutamate residues in an acidic pocket (references 31-34). However, it should be noted that in general, these other structures also possess a metal ion (sodium or calcium) and/or a basic sidechain (arginine or lysine) in the coordination sphere, forming a tight ion pair. Thus, the assigned bicarbonate binding site in GPR30 remains an anomaly in terms of the chemical properties of the proposed binding site.

Thank you for your insightful comments. Based on the weaknesses you pointed out, we reconstructed the receptor based on the improved density and removed the bicarbonate model. We performed calcium assays using cell lines stably expressing the variant based on the structure.

Reviewer #2(Public Review):

Summary:

In this manuscript, "Cryo-EM structure of the bicarbonate receptor GPR30," the authors aimed to enrich our understanding of the role of GPR30 in pH homeostasis by combining structural analysis with a receptor function assay. This work is a natural development and extension of their previous work (PMID: 38413581). In the current body of work, they solved the first cryo-EM structure of the human GPR30-G-protein (mini-Gsqi) complex in the presence of bicarbonate ions at 3.21 Å resolution. From the atomic model built based on this map, they observed the overall canonical architecture of class A GPCR and also identified 4 extracellular pockets created by extracellular loops (ECLs) (Pockets A-D). Based on the polarity, location, and charge of each pocket, the authors hypothesized that pocket D is a good candidate for the bicarbonate binding site. To verify their structural observation, on top of the 10 mutations they generated in the previous work, the authors introduced another 11 mutations to map out the essential residues for the bicarbonate response on hGPR30. In addition, the human GPR30-G-protein complex model also allowed the authors to untangle the G-protein coupling mechanism of this special class A GPCR that plays an important role in pH homeostasis.

Strengths:

As a continuation of their recent Nature Communication publication (PMID: 38413581), this study was carefully designed, and the authors used mutagenesis and functional studies to confirm their structural observations. This work provided high-resolution structural observations for the receptor in complex with G-protein, allowing us to explore its mechanism of action, and will further facilitate drug development targeting GPR30. There were 4 extracellular pockets created by ECLs (Pockets A-D). The authors were able to filter out 3 of them and identified that pocket D was a good candidate for the bicarbonate binding site based on the polarity, location, and charge of each pocket. From there, the authors identified the key residues on GPR30 for its interaction with the substrate, bicarbonate. Together with their previous work, they carefully mapped out nine amino acids that are critical for receptor reactivity.

Weaknesses:

It is unclear how novel the aspects presented in the new paper are compared to the most recent Nature Communications publication (PMID: 38413581). Some areas of the manuscript appear to be mixed with the previous publication. The work is still impactful to the field. The new and novel aspects of this manuscript could be better highlighted.

I also have some concerns about the TGF α shedding assay the authors used to verify their structural observation. I understand that this assay was also used in the authors' previous work published in Nature Communications. However, there are still several things in the current data that raised concerns:

Thank you for your insightful comments. Based on the weaknesses you pointed out, we highlighted the new and novel aspects of this manuscript could be better highlighted. We performed calcium assays using cell lines stably expressing the variant based on the structure.

(1) The authors confirmed the "similar expression levels of HA-tagged hGPR30" mutants by WB in Supplemental Figure 1A and B. However, compared to the hGPR30-HA (~6.5 when normalized to the housekeeping gene, Na-K-ATPase), several mutants of the key amino acids had much lower surface expression: S134A, D210A, C207A had ~50% reduction, D125A had ~30% reduction, and Q215A and P71A had ~20% reduction. This weakens the receptor reactivity measured by the TGF α shedding assay.

Since the calcium assay data is included in the main figure, the TGF α shedding assay and WB expression quantification data are Figure 3. — supplement figure 1-4, but we included an explanation of the expression levels in the figure caption.

(2) In the previous work, the authors demonstrated that hGPR30 signals through the Gq signaling pathway and can trigger calcium mobilization. Given that calcium mobilization is a more direct measurement for the downstream signaling of hGPR30 than the TGF α shedding assay, pairing the mutagenesis study with the calcium assay will be a better functional validation to confirm the disruption of bicarbonate signaling.

According to the suggestion, we performed calcium assay using cell lines stably expressing the mutants (Figure 3F, G and Figure 3—figure supplement 1-3).

(3) It was quite confusing for Figure 4B that all statistical analyses were done by comparing to the mock group. It would be clearer to compare the activity of the mutants to the wild-type cell line.

Thank you for your comment. As you mentioned, the comparisons are made between wild-type GPR30 and mutants in the revised manuscript (Figure 3G, Figure 3.—figure supplement

4B)

Additional concerns about the structural data include

(1) E218 was in close contact with bicarbonate in Figure 4D. However, there is no functional validation for this observation. Including the mutagenesis study of this site in the cell-based functional assay will strengthen this structural observation.

We cancelled the bicarbonate model, and we performed mutation analysis targeting all residues facing the binding pocket using cell lines that stably express variants including E218A.

(2) For the flow chart of the cryo-EM data processing in Supplemental data 2, the authors started with 10,148,422 particles after template picking, then had 441,348 Particles left after 2D classification/heterogenous refinement, and finally ended with 148,600 particles for the local refinement for the final map. There seems to be a lot of heterogeneity in this purified sample. GPCRs usually have flexible and dynamic loop regions, which explains the poor resolution of the ECLs in this case. Thus, a solid cell-based functional validation is a must to assign the bicarbonate binding pocket to support their hypothesis.

We re-analyzed the dataset and improved the local resolution of the extracellular region (Author response image 1) and cancelled the bicarbonate model. Yet, as suggested by the reviewer, solid cell-based functional validation is efficient to analyze the receptor function response to bicarbonate. Thus, we performed mutation analysis targeting all residues facing the binding pocket using cell lines stably expressing the mutants, whose surface expression levels were analyzed by FACS (Figure 3F, G and Figure 3.—figure supplement 1-3).

Reviewer #3 (Public Review):

Summary:

GPR30 responds to bicarbonate and regulates cellular responses to pH and ion homeostasis. However, it remains unclear how GPR30 recognizes bicarbonate ions. This paper presents the cryo-EM structure of GPR30 bound to a chimeric mini-Gq in the presence of bicarbonate. The structure together with functional studies aims to provide mechanistic insights into bicarbonate recognition and G protein coupling.

Strengths:

The authors performed comprehensive mutagenesis studies to map the possible binding site of bicarbonate.

Weaknesses:

Owing to the poor resolution of the structure, some structural findings may be overclaimed.

Based on EM maps shown in Figure 1a and Figure Supplement 2, densities for side chains in the receptor particularly in ECLs (around 4 Å) are poorly defined. At this resolution, it is unlikely to observe a disulfide bond (C130ECL1-C207ECL2) and bicarbonate ions. Moreover, the disulfide between ECL1 and ECL2 has not been observed in other GPCRs and the published structure of GPR30 (PMID: 38744981). The density of this disulfide bond could be noise.

The authors observed a weak density in pocket D, which is accounted for by the bicarbonate ions. This ion is mainly coordinated by Q215 and Q138. However, the Q215A mutation only reduced but not completely abolished bicarbonate response, and the author did not present the data of Q138A mutation. Therefore, Q215 and Q138 could not

be bicarbonate binding sites. While H307A completely abolished bicarbonate response, the authors proposed that this residue plays a structural role. Nevertheless, based on the structure, H307 is exposed and may be involved in binding bicarbonate. The assignment of bicarbonate in the structure is not supported by the data.

Thank you for your insightful comments. Based on the weaknesses you pointed out, we reconstructed the receptor based on the improved density and removed the bicarbonate model. We performed calcium assays using cell lines stably expressing the variant based on the structure.

Reviewer #1 (Recommendations For The Authors):

(1) The experimental validation of the bicarbonate binding could be strengthened by developing an assay that directly monitors bicarbonate binding (rather than GPCR signaling)

We agree that a direct binding assay for bicarbonate would be highly attractive (i.e. Filter binding assay using $^{14}\text{C-HCO}_3^-$). However, the weak affinity of bicarbonate ions (in the mM range) would make reliable radioisotope-based detection impossible due to minimal specific receptor occupancy and high non-specific background and thus it is highly challenging and there are limitations to what can be done in this structural paper.

and determining a structure at comparable resolution in the absence of bicarbonate. In addition, all residues that are proposed to be located adjacent to the bicarbonate should be mutated and functionally validated.

We re-modeled the receptor based on the improved density and canceled the bicarbonate model. We performed calcium assay using cell lines stably expressing the mutants (Figure 3F, G and Figure 3.-figure supplement 1-3).

(2) What are the maps contoured in Figure 4D? The legend should describe this. Is 218 within the map region shown, or is there no density for its sidechain?

We removed the corresponding figure and cancelled the bicarbonate model.

(3) The contour level of the maps in Figure 1 - Figure Supplement 2 should also be indicated. Are these all contoured at the same level?

Thank you for your comment. We re-analyzed the same data set and obtained new density maps and models. We reworked Figure 1 and Figure 1. figure supplement 2; the contour level of the map for Figure 1 and composite map for the Figure 1. figure supplement 2 is the same, 7.65.

(4) Regarding the cited structures of bicarbonate-binding proteins, for three of the four cited structures, the bicarbonate is actually coordinated by positive ligands, with the Asp/Glu playing a more peripheral role:

Capper et al: Overall basic cavity with tight bidentate coordination by Arg. The Glu is 5-6 Å away.

Koropatkin et al: Two structures. The first, solved at pH 5, is proposed to have carbonic acid bound. The second, solved at pH 8, shows carbonate in a complex with calcium, with the calcium coordinated by carboxylates.

Wang et al: The bicarbonate is coordinated by a lysine and a sodium ion. The sodium is coordinated by carboxylates.

The authors should more thoughtfully discuss the unusual properties of this binding site with regard to the previous literature. Is it possible that bicarbonate binds in complex

with a metal ion? Could this possibility be experimentally tested?

We cancelled the bicarbonate model.

(5) As a structure of GPR30 has been recently published by another group (PMID: 38744981), it would be valuable to discuss structural similarities and differences and discuss how bicarbonate activation and activation by the chloroquine ligand identified by the other group might both be accommodated by this structure.

Thank you for your valuable comment. We compared the structure presented by another group and added our discussion, as “During the revision of this manuscript, the structures of apo-GPR30-G_q (PDB 8XOG) and the exogenous ligand Lys05-bound GPR30-G_q (PDB 8XOF) were reported [42]. We compared our structure of GPR30 in the presence of bicarbonate with these structures. In the extracellular region, the position of TM5 in GPR30 in the presence of bicarbonate is similar to that in apo-GPR30. In contrast, the position of TM6 is shifted outward relative to that of apo-GPR30, resembling the conformation observed in Lys05-bound GPR30 (Figure 6A, B). Additionally, the position of ECL1 is also shifted outward compared to that of apo-GPR30 (Figure 6B). In the GPR30 structure in the presence of bicarbonate, ECL2 was modeled, suggesting differences in structural flexibility. These findings indicate that the structure of GPR30 in the presence of bicarbonate is different from both the apo structure and the Lys05-bound structure, demonstrating that the structure and the flexibility of the extracellular domain of GPR30 change depending on the type of ligand. Furthermore, focusing on the interaction with G_q, the αN helix of G_q is not rotated in the structure bound to Lys05, in contrast to the characteristic bending of the αN helix in our structure (Figure 6C, D). Although it is necessary to consider variations in experimental conditions, such as salt concentration, the differences in the G_q binding modes suggest that the downstream signals may change in a ligand-dependent manner.” (lines 249-266).

Reviewer #2 (Recommendations For The Authors):

(1) It is highly recommended that the authors carefully go through the "insights into bicarbonate binding" section. The results of the new findings in this paper were blended in with the results from the previous work: the importance of E115, Q138, and H307 in the receptor-bicarbonate interaction was shown in the Nature Communication paper but the authors didn't make it clear, which added a little confusion.

We emphasized this fact in the main text (lines 130-132).

(2) It would be nice for the authors to add some content about the physiological concentration of HCO₃ or refer more to their previous work about the rationale for selecting the bicarbonate dose in their functional assay.

Thank you for your comment. The physiological concentration of bicarbonate is 22-26 mM in the extracellular fluid, including interstitial fluid and blood, and 10-12 mM in the intracellular fluid. The bicarbonate concentration alters in various physiological and pathological conditions – metabolic acidosis in chronic kidney disease causes a drop to 2-3 mM, and metabolic alkalosis induced by severe vomiting increases HCO₃⁻ concentrations more than 30 mM. Thus, our present and previous works clearly show that GPR30 is activated by physiological concentrations of bicarbonate, whether it is localized intracellularly or on the membrane, and that GPR30 can be deactivated or reactivated in various pathophysiological conditions. We added this in the discussion section (lines 267-278).

(3) In Figure 3A, in the legend, the authors mentioned: "black dashed lines indicate hydrogen bonds". No hydrogen bond was noted in the figure.

We totally corrected Figure 3.

(4) Figure 3B, it would be helpful for the authors to denote the meaning of the blue-white-red color coding in the legend.

We removed the figure.

(5) Supplemental Figure 3: since AF3 was released on May 3rd, it would be awesome in the revision version if the authors would update this to the AF3 model.

The AF2 model has been replaced with the AF3. (Figure 2–figure supplement 2A-C). The AF2 and AF3 models are almost identical, and they form incorrect disulfide bonds. This confirms the usefulness of the experimental structural determination in this study.

(6) Supplemental Figure 4: it wasn't clear to me if the expression experiments were repeated multiple times or if there was any statistical analysis for the expression level was done in this study.

We performed the expression experiment by western blotting once and did not perform statistical analyses. We performed repeated FACS analyses of HEK cells stably expressing N-terminally HA-tagged wild-type or mutant GPR30s to analyze their membrane and whole-cell expressions during revision (Figure 3.–figure supplement 1-3). Using these stable cells, we performed calcium assays using cell lines stably expressing the mutants (Figure 3F, G and Figure 3–figure supplement 1-3).

(7) Supplemental Figure 4: Also, is there a reason for the authors to compare the expression level of hGPR30 to the housekeeping gene NA-K-ATPase rather than the total loaded protein? Traditionally housekeeping genes have been used as loading controls to semiquantitatively compare the expression of target proteins in western blots. However, numerous recent studies show that housekeeping proteins can be altered due to experimental conditions, biological variability across tissues, or pathologies. A consensus has developed for using total protein as the internal control for loading. An editorial from the Journal of Biological Chemistry reporting on "Principles and Guidelines for Reporting Preclinical Research" from the workshop held in June 2014 by the NIH Director's Office, Nature Publishing Group, and Science stated, "It is typically better to normalize Western blots using total protein loading as the denominator".

Thank you for your instructive comment. We evaluated western blotting with the same amount of total protein loaded 20 µg for whole-cell lysate and 1.5 µg for cell surface protein (Figure 3.–figure supplement 3C-F).

Reviewer #3 (Recommendations For The Authors):

The claim about this disulfide should be removed unless the authors can provide mass spec evidence.

Thank you for your crucial comments. Firstly, C130 is a residue of TM3, not ECL1, so our misprint has been corrected to C130^{3.25}. C207^{ECL2}, located at position 45.50, is the most conserved residue in ECL2, and it forms a disulfide bond with cysteine at position 3.25 (PMID: 35113559). The paper was additionally cited regarding the preservation of the bond of C130^{3.25}-C207^{ECL2} (line 103). Indeed, disruption of this disulfide bond by the C207^{ECL2} A mutation resulted in a marked reduction in receptor activity. In addition, the data set was re-analyzed to improve the local resolution of the extracellular region, and it was shown that the density of ECL2 is not noise (Figure 2. —figure supplement 2). We are confident about the presence of the disulfide bond, based on the structural analysis data and the conservation.

The highly flexible extracellular region is greatly affected by experimental conditions and ligands, so we speculate that the ECL2 and the disulfide bond was not observed in other reported structures of GPR30. Then, we have added the following content to the discussion, as

“In the GPR30 in the presence of bicarbonate, ECL2 was modelled, suggesting differences in structural flexibility.” (lines 256-257).

The authors should remove the assignment of bicarbonate in the structure, and tone down the binding site of bicarbonate.

We cancelled the bicarbonate model.

Minor:

(1) The potency of bicarbonate for GPR30 is in the mM range. Although the concentration of bicarbonate in the serum can reach mM range, how about its concentration in the tissues? Given its low potency, it may be not appropriate to claim GPR30 is a bicarbonate receptor at this point, but the authors can claim that GPR30 can be activated by or responds to bicarbonate.

The physiological concentration of bicarbonate is 22-26 mM in the extracellular fluid, including interstitial fluid and blood, and 10-12 mM in the intracellular fluid. Therefore, GPR30 is activated by physiological concentrations of bicarbonate in the tissues. Also, the bicarbonate concentration alters in various physiological and pathological conditions – metabolic acidosis in chronic kidney disease causes a drop to 2-3 mM, and metabolic alkalosis induced by severe vomiting increases HCO₃⁻ concentrations more than 30 mM. Thus, our work clearly shows that GPR30 is activated by physiological concentrations of bicarbonate, whether it is localized intracellularly or on the membrane, and that GPR30 can be deactivated or reactivated in various pathophysiological conditions. According to the reasons above, we claim GPR30 is a bicarbonate receptor (lines 267-278).

(2) The description that there is no consensus on a drug that targets GPR30 is not accurate, since lys05 has been reported as an agonist of GPR30 and their structure is published (PMID: 38744981). The published structures of GPR30 should be introduced in the paper.

We added the discussion about the structural comparison with the Lys05-bound structure (Figure 6, lines 249-266)

(3) BW numbers in Figure 4A should be shown.

We added BW numbers in the figures of the mutational studies.

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